

Re-introducing and testing the probabilistic sliding template model of vowel perception

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Abstract

Normalization of the speech signal onto comparatively invariant phonetic representations is critical to speech perception. Assumptions about this process also play a central role in phonetics and phonology. Yet, despite the importance of these assumptions, popular normalization accounts continue to incorrectly assume that the relevant normalization parameters are ‘known’—e.g., because the researcher can estimate them from a fully balanced set of recordings. Listeners, however, have to incrementally *infer* these parameters from the speech input. We re-introduce a seminal, but still underappreciated, model of this inference process—the Probabilistic Sliding Template Model of vowel normalization and perception (PSTM, Nearey and Assmann, 2007)—and show why researchers cannot afford to ignore it. We provide the R package STM, which makes it trivial to apply the PSTM to new data. We present the first quantitative assessment of the PSTM against human perception. We show that the model excel at predicting listeners’ vowel categorization behavior, *clearly* outperforming the popular Lobanov normalization.

1. Introduction

Cross-talker variability poses a computational challenge for speech perception: since different talkers can realize the same phonetic information with different acoustics, robust perception across talkers can only be achieved by adjusting to these differences. However, cross-talker differences in pronunciation also provide information about the language background and social identity of speakers. One way to approach these issues is to consider both linguistic and social perception as inference over the joint distribution of auditory percepts (e.g., in the form of phonetic cues), linguistic, and social categories. This can be conceptualized as similarity-based inferences over exemplars (Johnson 1997; Sumner 2011) or as Bayesian inferences (Kleinschmidt and Jaeger 2015; Kleinschmidt, Weatherholtz, and Jaeger 2018). However, there is of yet no widely available model that implements these ideas and makes them testable. This motivates the present work.

Here, we focus on vowel perception. We (re-)introduce the *Probabilistic Sliding Template Model (PSTM)* (Nearey and Assmann 2007) and—for the first time—evaluate it quantitatively in terms of its ability to explain human behavior. The PSTM is a Bayesian model of *vowel normalization*, the perceptual mapping from speech acoustic to a linguistic representation. The model is rooted in principled considerations about the biology of auditory perception in the mammalian brain, validated by cross-species comparisons. The formant normalization procedure proposed by (P)STM—*uniform scaling*, described in more detail below—is closely related to size variation due to the physics of acoustic resonators. Indeed, several animal species, such as koalas (Charlton et al. 2012), dogs (Taylor, Reby, and McComb 2010), and red deer (Reby et al. 2005), have been found to separate uniform

scaling (size information) from the communicative information present in the signal (for review, Barreda 2020).

A major appeal of the PSTM is that it integrates social and linguistic inferences into a joint inference process, simultaneously estimating speaker characteristics and categorizing vowels. This allows for the empirical investigation of the connection between speech perception, social knowledge, and the estimation of speaker indexical characteristics. Further, unlike most existing models of normalization, the PSTM infers the speaker characteristics required for normalization *incrementally* from the speech input (via one-shot inferences from a single vowel input). As we show below, it does so with high accuracy. This helps address the ‘bootstrap’ problem of speech normalization: listeners must know some speaker characteristics in order to normalize (and identify) speech but must also identify speech in order to accurately estimate these same characteristics. Speaker characteristics and speech acoustics enter into the same many-to-many mapping relationship as the linguistic signal and speech acoustics. Consequently, determining both simultaneously from the speech signal is difficult, and the PSTM is one of the few models that addresses this problem in an empirically testable manner.

The incremental inference mechanism described by the PSTM stands out even when compared to modern models of automatic speech recognition (ASR; e.g., HuBERT, Wav2vec, Whisper). These models have made tremendous progress in understanding human speech even under challenging conditions (Kim et al. 2024). Recent work further suggests that the perceptual representations learned by these ASR models capture some aspects of human speech perception (Chernyak et al. 2024; Jin, Zhu, and Jaeger 2025). Unlike the PSTM, however, none of these models exhibits the rapid, exposure-driven, incremental changes in perception that human listeners exhibit (Magnuson and Nusbaum 2007; Barreda 2012; Persson, Barreda, and Jaeger 2025).

1.1 Why we wrote this paper, and why you might want to read it

We believe that Nearey and Assmann (2007) is one of the most under-appreciated papers on speech perception. It has been cited less than 40 times over 17 years. During the same time, there have been over 17,000 research articles on vowel identification and formant normalization. Among the articles covering formant normalization, Lobanov normalization (Lobanov 1971) is cited about 100-times more often than the PSTM (Nearey and Assmann 2007)—despite the fact that the latter is the only published model of the incremental inferences underlying vowel perception. So, here is a ground-breaking idea and promising candidate model of how humans perceive spectral information, with far-reaching consequences for research from typology to sociolinguistics to speech perception, ... and few seem to know of it.

We see two reasons for this. First, the PSTM has not previously been tested against human perception. Nearey and Assmann (2007) evaluated the PSTM in terms of its ability to recognize the vowel *intended by the talker*. However, models of human speech perception should be evaluated in terms of their ability to predict *what listeners perceive* (see also

Barreda 2021; Persson, Barreda, and Jaeger 2025). We provide such a test below, and demonstrate that the PSTM is a far superior model of human formant perception than the most-cited normalization account.

Second, in an effort to make the PSTM more accessible, we re-introduce it along with helpful visualizations. Our supplementary information (SI) provides step-by-step derivations of the PSTM. We have also implemented the PSTM in the R package STM, which is designed to be user-friendly. The present article serves as a demonstration: it is written in Quarto markdown and uses the STM package. Other researchers can download this markdown at <https://osf.io/tpwmv/>, and re-run our analyses with the press of a button in RStudio (R Core Team 2023; RStudio Team 2024), or apply similar analyses to their own data.

2. Perceptual normalization through uniform scaling

A core assumption of normalization accounts is that dialect-specific vowel representations are learned and represented in a *normalized* formant space. We start by motivating the normalized formant space assumed by the (P)STM.

Formant inputs from different talkers tend to be perceived as phonetically similar if they differ only according to a single multiplicative scaling parameter (for review, see Barreda 2020). For example, in order to preserve phonetic similarity, a speaker who produces an F1 that is 10% higher for /a/ than another should also produce F2 and F3 that are 10% higher—i.e., all formants are *uniformly scaled*. Consider a dialect-specific formant target $\vec{F}_v^* := [F1_v^*, F2_v^*, F3_v^*, \dots]$ for vowel v . A speaker s of that dialect, should target formants $\vec{F}_v^{s*} := [F1_v^{s*}, F2_v^{s*}, F3_v^{s*}, \dots]$ that are *uniformly scaled* by a speaker-specific scaling parameter ρ_s , as in Equation 1.

$$\vec{F}_v^{s*} := [F1_v^{s*}, F2_v^{s*}, F3_v^{s*}, \dots] = [F1_v^*, F2_v^*, F3_v^*, \dots] \cdot \rho_s \quad (1)$$

Equation 1 can be re-expressed as the sum of log-transformed formant targets $\vec{G}_v^* := [G1_v^*, G2_v^*, G3_v^*] = \log(\vec{F}_v^*)$, where $\psi_s = \log(\rho_s)$, as in Equation 2. In log-transformed Hz, uniform scaling thus results in additive, rather than multiplicative, changes. This relation between multiplicative scaling in Hz and additive scaling in log-Hz is illustrated in Figure 1.

$$\vec{G}_v^{s*} := [G1_v^{s*}, G2_v^{s*}, G3_v^{s*}, \dots] = [G1_v^*, G2_v^*, G3_v^*, \dots] + [\psi_s, \psi_s, \psi_s, \dots] \quad (2)$$

most commonly used normalization method in speech research—Lobanov normalization—assumes that listeners infer and store two parameters *per feature*. For instances, to model normalization of F1-F3 at two time points in the vowel (as we do below), listeners would need to estimate twelve parameters. This additional complexity of Lobanov normalization is one of the reasons it might not be an adequate model for human speech perception (see discussion in Barreda 2020, 2021; Persson, Barreda, and Jaeger 2025).

3. The Probabilistic Sliding Template Model of vowel perception (PSTM)

Nearey and Assmann (2007) referred to listeners’ dialect-specific implicit knowledge about each vowel’s formant distributions as “probabilistic templates”, where listeners are assumed to learn dialect-specific vowel representations over the *normalized* vowel formants. These templates could be represented, for instance, as exemplar clouds (Johnson 1997) or as parametric distributions (Nearey and Assmann 2007). For computational tractability, we follow the latter approach and represent vowel categories as multivariate Normal distributions over normalized formants, with mean vectors $\vec{\mu}_v$ and covariance matrices $\vec{\Sigma}_v$. We may then use the multivariate normal density (*MVN*) of each vowel to calculate the likelihood of any observed log-transformed formant input $\vec{G}$ given that vowel and the speaker-specific scaling parameter ψ_s . This can be done by either sliding the template as in Figure 1(d) or, equivalently, by scaling the formants into the normalized template space:

$$P(\vec{G}|\mathbf{v}, \psi_s) := MVN(\vec{G}|\vec{\mu}_v + \vec{\psi}_s, \vec{\Sigma}_v) = MVN(\vec{G} - \vec{\psi}_s|\vec{\mu}_v, \vec{\Sigma}_v) = MVN(\vec{N}|\vec{\mu}_v, \vec{\Sigma}_v) \quad (4)$$

The likelihoods in Equation 4 can be combined with the prior probability of each vowel category in the current context to find the *posterior probability* of each vowel category $\mathbf{v}_i$, as in Equation 5 (V is the number of unique vowel categories).

$$P(\mathbf{v}_i|\vec{G}, \psi_s) = \frac{P(\vec{G}|\mathbf{v}_i, \psi_s) \cdot P(\mathbf{v}_i)}{\sum_{j=1}^V P(\vec{G}|\mathbf{v}_j, \psi_s) \cdot P(\mathbf{v}_j)} \quad (5)$$

This posterior probability provides a gradient measure of category membership (Nearey and Assmann 2007; Luce and Pisoni 1998; Norris and McQueen 2008; Xie, Jaeger, and Kurumada 2023)—conceptually paralleling the gradient activation of categories in connectionist or exemplar models. Listeners’ responses in an n -alternative forced-choice categorization task can be modeled via decision rules based on the posterior probability. For instance, listener might always respond with the vowel that has the highest posterior probability (as in Nearey and Assmann 2007) or respond by sampling from the posterior (Luce’s choice rule, Massaro and Friedman 1990).

Given an estimate of the speaker-specific scaling parameter ψ_s , listeners can thus categorize formant inputs under any dialect-specific template. Similarly, researchers can use speaker-specific ψ_s estimates to project formant data from various talkers into a common normalized space. This can be useful when comparing vowel productions across different types of speakers—for instance, to assess effects of social identity, language background, or to compare clinical and neurotypical populations. However, a speaker’s ψ_s is a latent variable not directly present in their speech. As a result, both researchers and listeners must *estimate* ψ_s , as we describe next.

4. Probabilistic estimation of ψ_s

Researchers interested in projecting formant estimates $\vec{F}$ from different talkers into a common normalized space might be tempted to employ a seemingly straightforward approach to estimate of the speaker-specific $\hat{\psi}_s$ —simply average all log-transformed formant inputs $\vec{G}$ from the speaker (method 1 from Nearey and Assmann 2007):

$$\hat{\psi}_s := \bar{G} = \frac{1}{(K \cdot N)} \sum_{j=1}^N \sum_{k=1}^K G_{j,k,s} \quad (6)$$

where N is the number of observations per talker, K is the number of formants per input (e.g. $K = 2$ if only F1 and F2 are considered), and $G_{j,k,s}$ is the k th formant of the j vowel observation of speaker s . Under this formulation, uniform scaling is a form of extrinsic normalization, relying on information that is collected across observations.

However, this approach quickly becomes unfeasible even for researchers. First, estimates based on Equation 6 can be highly sensitive to formant measurement errors when the number of recordings per vowel is small. Second, even for large data sets, the approach in Equation 6 runs into substantial problems when comparing within or across data sets with different numbers of observations per vowel and talker (for discussion, Barreda and Nearey 2018; Nearey and Assmann 2007; Xie, Jaeger, and Kurumada 2023, SI 2.1). This problem arises for the same reasons that make formants relevant to vowel perception in the first place: the distribution of formants—and thus their mean—differs between vowels. This means that Equation 6 will yield systematically different estimates of $\hat{\psi}_s$, even for the same speaker with the same underlying ψ_s , depending on the specific set of vowel instances over which ψ_s is estimated.

The same considerations make the approach in Equation 6 unsuitable for *listeners*. And, unlike researchers, listeners who encounter an unfamiliar talker do not have the luxury of waiting until they have observed a large amount of formant inputs from that talker: if normalization via uniform scaling underlies robust speech perception, listeners must quickly arrive at adequate estimates of ψ_s —ideally given a single instance of a vowel that they hear from the unfamiliar talker. If listeners were using something like Equation 6 to estimate ψ_s for individual vowel tokens, their estimates of ψ_s would vary substantially from

trial to trial—depending also on which vowels the talker has produced. This would result in both highly inaccurate speech perception and very large changes to inferred speaker characteristics that themselves depend on ψ_s (e.g., size and gender). Neither of these outcomes are observed in the literature.

Nearey and Assmann (2007) addressed this issue by proposing several probabilistic approaches to ψ_s estimation that do not rely on a balanced sample of the speaker’s entire vowel system. These models share two key methodological insights. First, they constrain estimates of ψ_s given the listener’s assumed phonological knowledge (i.e. the template). Second, listeners are assumed to have prior expectations about the distribution of ψ_s depending on the acoustic and indexical characteristics of the talker. Here, we focus on three of the methods proposed by Nearey and Assmann (methods 2, 3, and 6), differing only in their assumptions about listeners’ prior expectations about ψ_s . Specifically, the three methods make increasingly stronger assumptions about the implicit knowledge that listeners have learned and stored about the distribution of ψ_s . By comparing how well these methods fit listeners’ behavior, one can therefore test hypotheses about the type of implicit knowledge listeners have. Before we turn to the differences between the three methods, we describe their shared characteristics.

4.1 Estimating ψ_s using prior expectations about vowel templates and ψ_s

Methods 2, 3, and 6 share that they estimate the *maximum a posteriori* (MAP) value of ψ_s , as shown below in (Equation 10). Since Nearey and Assmann (2007) did not provide the derivation of these MAP estimates, we briefly summarize the core insights. We start by using Bayes theorem to calculate the posterior distribution of ψ_s given the observed formant pattern, the vowel category, and the listener’s prior expectations about the distributions of ψ_s and vowel categories:¹

$$P(\psi_s | \vec{G}, v_j) = \frac{P(\vec{G} | v_j, \psi_s) \cdot P(\psi_s) \cdot P(v_j)}{\sum_{i=1}^V P(\vec{G} | v_i, \psi_s) \cdot P(\psi_s) \cdot P(v_i)} \quad (7)$$

As per Equation 4, the likelihood $P(\vec{G} | v, \psi_s)$ given the vowel category and ψ_s is assumed to be a multivariate normal distribution. Researchers can estimate the mean $\vec{\mu}_v$ and covariance matrix $\vec{\Sigma}_v$ for each vowel category based on any reasonably sized database of normalized vowel formants from the dialect(s) that the listener is assumed to have learned their template(s) from. Similarly, $P(v)$, can be calculated from relevant speech corpora or, for many experimental contexts, assumed to be equal across all categories and ignored in the calculation.

¹ We assume independence of $P(\psi_s)$ and $P(v)_j$ since the prior distribution over ψ_s is unlikely to depend on the prior probability of the different vowels in the current context.

The posterior distribution in Equation 7 can be used to obtain separate MAP estimates of ψ_s for each vowel category, $\hat{\psi}_{s,v}$. A vowel system with V vowels would result in V MAP estimates $\hat{\psi}_{s,v}$:

$$\begin{aligned}\hat{\psi}_{s,v=1} &:= \underset{\psi_s}{\operatorname{argmax}} [P(\vec{G}|\mathbf{v}=1, \psi_s) \cdot P(\psi_s) \cdot P(\mathbf{v} = 1)] \\ \hat{\psi}_{s,v=2} &:= \underset{\psi_s}{\operatorname{argmax}} [P(\vec{G}|\mathbf{v}=2, \psi_s) \cdot P(\psi_s) \cdot P(\mathbf{v} = 2)] \\ &\vdots \\ \hat{\psi}_{s,v=V} &:= \underset{\psi_s}{\operatorname{argmax}} [P(\vec{G}|\mathbf{v}=V, \psi_s) \cdot P(\psi_s) \cdot P(\mathbf{v} = V)]\end{aligned}\quad (8)$$

Nearey and Assmann (2007) use the best $\hat{\psi}_{s,v}$ for each vowel to calculate the posterior probability of each category given the formant input, as in Equation 9. Note that this is an update of Equation 5 using vowel-specific estimates of ψ_s , and the prior probability of that estimate.

$$P(\mathbf{v}_j|\vec{G}, \hat{\psi}_{s,v=j}) = \frac{P(\vec{G}|\mathbf{v}_j, \hat{\psi}_{s,v=j}) \cdot P(\psi_s = \hat{\psi}_{s,v=j}) \cdot P(\mathbf{v}_j)}{\sum_{i=1}^V P(\vec{G}|\mathbf{v}_i, \hat{\psi}_{s,v=i}) \cdot P(\psi_s = \hat{\psi}_{s,v=i}) \cdot P(\mathbf{v}_i)} \quad (9)$$

These $\hat{\psi}_{s,v}$ are then used to categorize the formant input into a winning category, c . Specifically, listeners are assumed to infer the $\hat{\psi}_s$ with the highest MAP, and to use this $\hat{\psi}_s$ to categorize the input as the vowel category with the maximum posterior density (Equation 10). Nearey and Assmann describe this process as “choose the vowel that looks best when it tries to look its best” (p. 253).

$$\begin{aligned}c &:= \underset{\mathbf{v}}{\operatorname{argmax}} [P(\mathbf{v}|\vec{G}, \hat{\psi}_{s,v})] \\ \hat{\psi}_s &:= \underset{\psi_s}{\operatorname{argmax}} [P(\psi_s|\vec{G}, \mathbf{v} = c)] = \hat{\psi}_{s,v=c}\end{aligned}\quad (10)$$

4.2 Method 2: Unrestricted optimization of ψ_s

Method 2 spells out Equation 7 as Equation 11, assuming that listeners’ prior expectations about ψ_s correspond to a uniform distribution (U) such that any value of ψ_s is equally plausible a priori.

$$\begin{aligned}\hat{\psi}_{s,v} &:= \underset{\psi_s}{\operatorname{argmax}} [P(\vec{G}|\mathbf{v}, \psi_s) \cdot P(\psi_s) \cdot P(\mathbf{v})] \\ &:= \underset{\psi_s}{\operatorname{argmax}} [P(\vec{G}|\mathbf{v}, \psi_s) \cdot U(\psi_s) \cdot P(\mathbf{v})]\end{aligned}\quad (11)$$

Using a uniform prior on ψ_s means that our ‘best’ estimate of ψ_s can be rather implausible, as long as it provides a good fit to one of our categories. For example, Figure 2 presents the likelihood functions $P(\vec{G}_x|\mathbf{v}_i, \hat{\psi}_s)$ from Equation 11 for four vowel hypotheses, along with different interpretations of an ambiguous vowel according MAP estimates $\hat{\psi}_{s,v}$ for the four vowels. With the uniform prior, the likelihoods and posteriors share the same relative

shapes in this case. As a result, method 2 suggests that /o/, /ɑ/, and /ʌ/ are all among the probable solutions. However, some of these vowels would result in estimates of ψ that are, as we will see, *a priori* highly improbable (e.g., for /o/, $\hat{\psi}_s = 7.64$). Method 3 takes this into account.

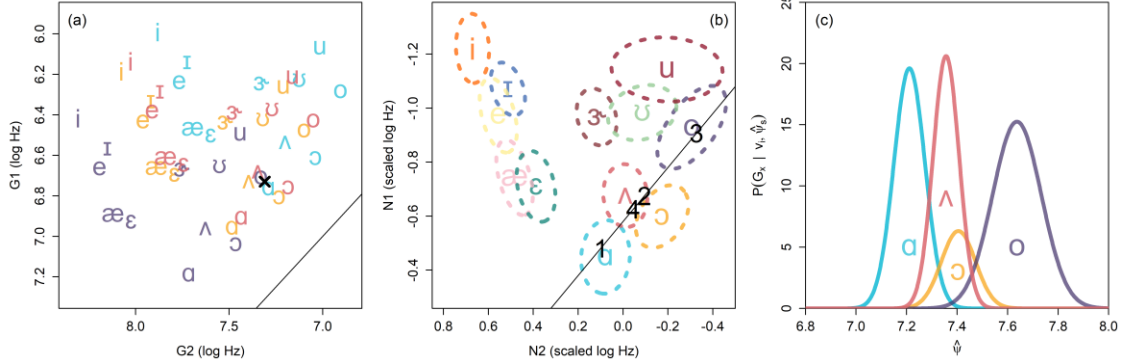


Figure 2: (a) A formant input “x”, relative to four possible positions of the dialect template, corresponding to four different values of $\hat{\psi}_s$. The template ‘slides’ along $G1 = G2$ (black line). (b) Alternatively, the vowel can be thought of ‘sliding’ across the normalized template. Black line indicates possible interpretations of the point in (a), ellipses enclose one standard deviation. Points indicate the best possible locations for /ɑ/ (1), /c/ (2), /o/ (3), and /ʌ/ (4), according to method 2. These interpretations correspond to the different vowel spaces (colors) in (a). (c) Posterior distributions of ψ_s given different vowel categories. Because method 2 uses a uniform prior, these posteriors are proportional to the likelihood. Thus, curves in (c) represent the vowel-specific densities of $\vec{G}_x$, $P(\vec{G}_x | v_i, \hat{\psi}_s)$, along the black line in (b). This highlights the relationship between categorization and ψ_s estimation.

4.3 Method 3: Informative expectations about ψ_s

Method 3 (Equation 12) introduces stronger constraints on listeners’ prior expectations for ψ_s by assuming that ψ_s is normally distributed across speakers, with a mean μ_{ψ_s} and standard deviation σ_{ψ_s} based on values the listener previously observed across speakers. This means that values of $\hat{\psi}_{s,v}$ that are closer to the population mean will be considered more generally plausible by that listener.

$$\begin{aligned} \hat{\psi}_{s,v} &:= \underset{\psi_s}{\operatorname{argmax}} [P(\vec{G} | v, \psi_s) \cdot P(\psi_s) \cdot P(v)] \\ &:= \underset{\psi_s}{\operatorname{argmax}} [P(\vec{G} | v, \psi_s) \cdot \text{Normal}(\psi_s | \hat{\mu}_{\psi_s}, \hat{\sigma}_{\psi_s}) \cdot P(v)] \end{aligned} \quad (12)$$

Just like the multivariate Normal distributions for the vowel templates, estimates of μ_{ψ_s} and σ_{ψ_s} can be obtained from a reasonably sized database. For instance, Nearey and Assmann (2007) suggest $\hat{\mu}_{\psi_s} = 7.23$ and $\hat{\sigma}_{\psi_s} = 0.128$ based on a database of 265 speakers of US English (86 adult females, 88 adult males, 91 children) from three dialect regions

(Hillenbrand et al. 1995; Peterson and Barney 1952; Assmann and Katz 2000).² The two leftmost columns of Figure 3 compare methods 2 and 3, showing that posterior probabilities can be affected even by comparatively weak constraints on the prior of ψ_s . For instance, the vowel /o/ and the $\hat{\psi}_s$ values it would entail are no longer probable since those ψ_s values are *a priori* improbable (e.g., $\psi_s = 7.64$ is more than 3 standard deviations above the mean of the prior distribution).

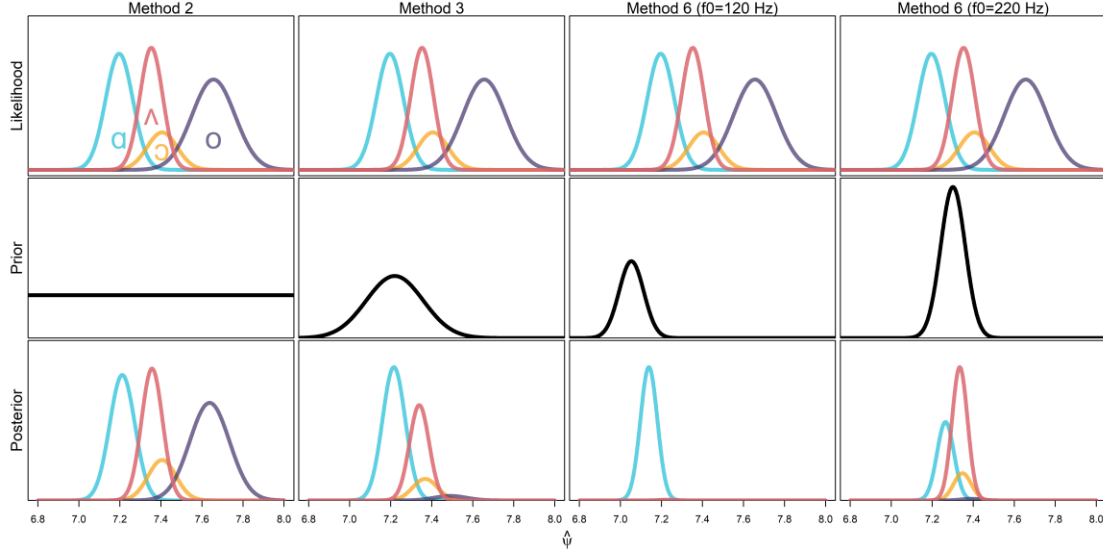


Figure 3: Comparison of different approaches (columns) to estimating prior probability of ψ_s . (top) Likelihood of ψ_s values given different vowel categories in Figure 2. (middle) Prior distribution of ψ_s . (bottom) Posterior probabilities of ψ_s conditional on vowel category (combining likelihood and prior in each column). The most probable $\hat{\psi}_s$ (location on x-axis of highest peak), and vowel category (color of curve with highest peak), changes as a function of selected prior, and vowel token's f_0 .

4.4 Method 6: Informative expectations about ψ_s conditional on f_0

Method 6 further strengthens the constraints on listeners' prior expectations for ψ_s by conditioning those expectations on talkers' f_0 . Specifically, method 6 finds the MAP estimate of ψ_s while considering both the likelihood of the observed log-transformed formant input $\vec{G}$ and the likelihood of the observed $g_0 := \log(f_0)$, as in Equation 13.

$$\hat{\psi}_{s,v} := \operatorname{argmax}_{\psi_s} [P(\vec{G}|v, \psi_s) \cdot P(g_0|\psi_s) \cdot P(\psi_s) \cdot P(v)] \quad (13)$$

Just like the mean and standard deviation of the normal prior for method 3, researchers can estimate $P(g_0|\psi_s)$ in Equation 13 from a reasonably-sized database. Specifically,

² The STM package allows researchers to use other data sources to replace or refine these estimates.

assuming a linear relationship between $g0$ and ψ_s with normally distributed residual uncertainty about $g0$, researchers can use linear regression to predict observed $g0_s$ from estimates of $\hat{\psi}_s = \bar{G}_s$ as in (Equation 6).³ The intercept $\hat{\alpha}_{g0}$ and slope $\hat{\beta}_{g0}$ can then be used to predict the mean $\hat{\mu}_{g0}$ around which $g0$ is expected to be distributed given ψ_s with a standard deviation equal to the residual standard deviation of the linear regression, $\hat{\sigma}_{g0}$:

$$P(g0|\psi_s) := N(\hat{\mu}_{g0}, \hat{\sigma}_{g0}) = N(\hat{\alpha}_{g0} + \hat{\beta}_{g0} \cdot \psi_s, \hat{\sigma}_{g0}) \quad (14)$$

Using the same database from method 3, Nearey and Assmann obtained $\hat{\alpha}_{g0} = -10.3$, $\hat{\beta}_{g0} = 2.14$, and $\hat{\sigma}_{g0} = 0.133$. All other steps required to obtain the MAP estimate of ψ_s parallel method 3. The two rightmost columns of Figure 3 illustrate how $f0$ can affect both the estimation of ψ_s and the posterior probabilities of different vowel categories.

4.5 Implementation

Nearey and Assmann (2007) note that “analytic solutions to the optimizations [for the different methods] are available and no search is necessary” (p. 252). However, they did not provide those analytic solutions. In Equation 15, we state the analytic solutions for method 6, which contains all calculations required for methods 2 and 3. To find the MAP estimates $\hat{\psi}_{s,v}$ for each vowel, we must calculate the product of the densities in Equation 15 for each vowel and find the maximum. The complete derivation is provided in the SI.

$$\begin{aligned} P(\vec{G}|\mathbf{v}, \psi_s) \cdot P(g0|\psi_s) \cdot P(\psi_s) \cdot P(\mathbf{v}) = \\ \text{MVN}(\vec{N}_\psi | \vec{\mu}_\psi, \hat{\Sigma}_\psi) \cdot \\ N(\hat{\alpha}_{g0} + \hat{\beta}_{g0} \cdot \psi_s, \hat{\sigma}_{g0}) \cdot \quad (15) \\ N(\hat{\mu}_\psi, \hat{\sigma}_\psi) \cdot \\ P(\mathbf{v}) \end{aligned}$$

5. Bootstrap Evaluation

To compare the performance of different implementations of the STM in understanding perceptual vowel normalization, we conduct a bootstrap analysis using data on the perception of US English vowels (Hillenbrand et al. 1995). A walk-through of the analysis is available as an R markdown vignette in the OSF repo listed above. The STM package allows researchers to conduct similar analyses on their own data, with just a handful of R

³ The database used for this purpose should have formant and $f0$ measurements for all vowels, and should be balanced with equally many instances of each vowel both within and across recorded talkers. Otherwise the mean of the log-formants, $\bar{G}_s$, will be systematically biased by differences in the distribution of vowel instances across talkers. See Barreda and Nearey (2018) for strategies when data is not balanced.

commands. Details regarding the bootstrap evaluation are found in the Appendix. For each bootstrap data set, we compare approaches based on three performance metrics:

- A) The **likelihood of listeners' categorization responses** $\Lambda := \sum_{i=1}^m \sum_{j=1}^n C_{ij} \cdot \log(P_{ij})$, where C_{ij} is the number of times token i was classified as vowel category j , and $\log(P_{ij})$ is the log-transformed posterior probability for token i and category j . This is the critical measure of how well a method describes listeners' categorization behavior, which has not previously been provided.
- B) The **likelihood of the vowel category that the talker intended to produce** (or, specifically, that the experimenter asked the talker to produce). This metric is closer to what a speech engineer would choose to evaluate normalization methods, as it measures how well the method performs in recognizing the *intended* vowel category. While this metric does *not* assess how well a method explains speech perception, it provides an important comparison, as we explain below.
- C) The **root-mean square (RMS) error in estimating ψ_s** compared to an estimate of $\hat{\psi}$ obtained from a balanced sample of each speaker's entire vowel system, and [Equation 6](#). Ultimately, researchers interested in assessing how well a method describes listeners' estimation of ψ_s should compare each method's predictions for $\hat{\psi}_s$ against listeners' estimates.

Of note, the use of likelihood as a metric to assess A) the fit against listeners' perception and B) the intended category differs from the approach in Nearey and Assmann (2007). Nearey and Assmann used the accuracy (under the criterion choice rule) of predicting the intended category label for B) and, critically, did not evaluate against listeners' perception. The use of accuracy as a metric is now understood to be problematic, as it does not capture how well a model predicts the *distribution* of responses (for discussion, Persson, Barreda, and Jaeger 2025).

5.1 Results

The results of the bootstrap analysis are presented in [Figure 4](#). Method 6 offers the best fit against listeners' categorization responses for the Hillenbrand et al. (1995) data. Based on the bootstrap, all differences—except for the contrast between method 6 and unnormalized Hz—are significant, i.e., 95% quantiles do not overlap 0 in [Figure 4 \(a\)](#). However, method 6 performed better than unnormalized Hz in 94.4% of all bootstrap iterations. This shows the importance of considering prior information about ψ_s during speech perception, and also the potential usefulness of considering the joint distribution of ψ_s and f_0 in perception.

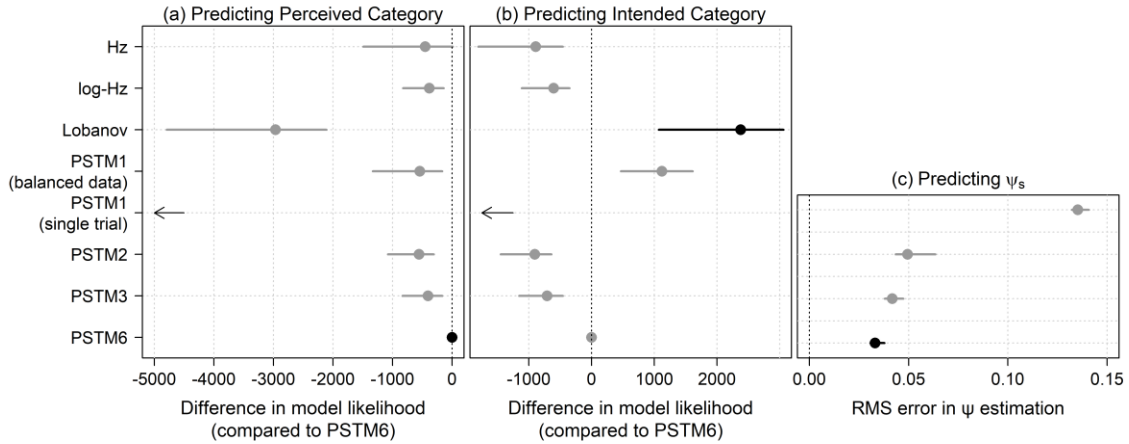


Figure 4: (a) Difference in model log-likelihoods of listeners’ responses between each model and PSTM method 6 across bootstrap iterations. Point ranges show bootstrap means and lower 95% quantiles. Left-arrows indicate very low log-likelihoods for single-trial application of method 1 (more than one order of magnitude lower than all other likelihoods). (b) Same as (a) but for the vowel category that the talker intended to produce, rather than what listeners responded. (c) Root-mean-squared (RMS) error for ψ_s prediction for the approaches that estimate ψ_s (only methods that provide estimates of ψ_s are shown).

Of particular note, method 6 provides a better fit to listeners’ responses than the ‘classic’ approach over balanced data (method 1 in Nearey and Assmann 2007). This is of note because the classic approach—which estimates ψ_s from the balanced and complete training data—has access to far more speaker-specific information than method 6. As a consequence, the classic approach achieves better performance in predicting the vowel category that the *talker intended to produce* (see Figure 4(b)). Yet, the classic approach does *not* provide the best explanation for *listeners’* categorization responses. This highlights what should be obvious—and yet is often ignored in research on normalization: that listeners do not have access to a large and balanced set of vowel productions from a talker when they first encounter them. Effective speech perception thus requires mechanisms that can incrementally infer the relevant quantities (here ψ_s) based on prior expectations and other available information (e.g., f_0 , as in method 6).

Just how well method 6 performs is further highlighted by the comparison against the normalization approach that arguably remains the most popular, Lobanov normalization. While Lobanov normalization provides the best fit to the intended vowel category, it is one of the worst models of listener responses. This lends additional emphasis to a point we have made elsewhere: just because a normalization procedure excels at reducing cross-talker formant variability this does *not* mean it is an adequate model of human perception (e.g., Barreda 2021; Persson, Barreda, and Jaeger 2025).

Method 6 also outperforms method 1 when the latter is applied trial-by-trial (see appendix). We included this variant of method 1 (not considered in Nearey and Assmann

2007) as a more direct baseline to the single-trial methods 2, 3, and 6. The fact that the single-trial variant of method 1 performs so poorly that it is not even visible in [Figure 4](#) (a) validates the issues we raised in [Section 4](#) about method 1: it would lead to highly unstable vowel perception, and listeners do not seem to use it.

Finally, another appealing feature of the STM is that it provides estimates of ψ_s ‘for free’, naturally tying together speech perception and the perception of speaker size. Since we do not know listeners’ estimates of ψ_s , we cannot directly compare the different methods against that ground truth. We can, however, compare how accurately the incremental PSTM methods estimate the ψ_s that would result from calculating ψ_s over the entire data from the talker (our best estimate of ψ_s). [Figure 4](#) (c) shows that method 6 again performed best (lowest RMS errors).

6. Conclusions and Future Directions

Almost two decades after its introduction, the PSTM remains the only published, fully spelled-out model of incremental formant normalization and vowel perception. By making the code for this model available in the form of an R package, we hope to make the advantages of the PSTM and its extensions more accessible to other researchers. The PSTM’s approach to normalization, uniform scaling, has been consistently found to provide a better fit against listeners’ perception than other influential methods, such as Lobanov normalization (Barreda 2021; Persson, Barreda, and Jaeger 2025; Richter et al. 2017).

Here, we show that the predictive power of the PSTM improves further, and in non-subtle ways, once the *incremental* inferences listeners need to make are adequately modeled. Our bootstrap simulations confirm that method 6—which assumes that listeners consider both intrinsic and extrinsic information in guiding their prior expectations about ψ_s —best explains listeners’ behavior in the data from Hillenbrand et al. (1995).

The general framework for incremental normalization presented here can be extended to a wide range of other normalization methods. This will allow comparisons between different approaches to incremental formant normalization and beyond. In particular, future work can investigate how incremental normalization can be integrated into modern ASR architectures, to improve the suitability of these architectures as a model of human speech perception. This will also help researchers move beyond normalization of formant point estimates (e.g., F1-F3 at the start and end of the vowel), addressing longstanding concerns that human listeners use a much broader sets of acoustic features when identifying vowels Barreda and Nearey (2012).

7. Appendix: Bootstrap details

The Hillenbrand et al. (1995) data include acoustic measures from 139 speakers producing one instance of twelve vowels each, including formant measures at multiple time slices.

We used formant measurements at 20% and 80% of the vowel duration. This meant each vowel token was represented by a vector of length six, two measurements for each of three formants. Following Nearey and Assmann (2007), we used two time points, capturing that formant dynamics affect vowel perception for the dialect in the database (Hillenbrand and Nearey 1999). By fitting multivariate normal distributions in this six-dimensional space, we assume that listeners have learned expectations about the (co)variance between formants within and across time points.

The Hillenbrand et al. (1995) data also contain 12-alternative forced-choice categorization responses for each vowel token, aggregated across 20 listeners of the same dialect as the speakers (average accuracy = 95%).⁴ Crucially, these responses were elicited over stimuli from the different speakers in the database, presented in randomized order—i.e., precisely the type of input for which the naive estimation of ψ_s (Equation 6) is expected to yield unreliable results for listeners (confirmed below).

We bootstrapped the following process over 1000 iterations for each method (all functions referred to are from the STM package):

1. Randomly divide the data into a 79-speaker training set and a 60-speaker testing set. Resample the training data (with replacement) at the speaker level. Sixty-three vowel tokens (4%) had one or two missing formant measurements (out of 6), representing 0.09% of the total formant measurements in the data. The missing values were imputed using stochastic regression imputation, independently for each iteration (using the `impute_NA` function).
2. Uniformly scale the training data (the `normalize` function), by estimating each speaker's ψ_s using their complete vowel system as in Equation 6, and normalizing it as in Equation 3.
3. Estimate the dialect-specific category means and covariance matrices of all vowels using the normalized training data (with `create_template`).⁵ **Each iteration of the bootstrap simulates a ‘listener’ of the dialect with slightly different speech experience.**

⁴ The data used to estimate vowel templates should reflect the sort of listener that the researcher aims to model. E.g., a template trained on a database of Canadian English will not be suitable to model the perception of an Irish English listener (see discussion in Persson, Barreda, and Jaeger 2025).

⁵ We follow Nearey and Assmann (2007) in estimating a single pooled covariance matrix across all vowel categories. It is an open question whether category-specific covariance matrices, as in Equation 4, provide a better fit to listener's responses. The STM package supports either option.

4. For each token in the testing data, estimate $\hat{\psi}_{s,v}$ for each vowel category using each method. Use these values of $\hat{\psi}_{s,v}$ to obtain the winning estimate $\hat{\psi}_s$ and calculate the posterior probability of each vowel category.
5. Calculate performance metrics A)-C) described in [Section 5](#).

We compare eight different methods that differ in terms of whether and/or how they estimate ψ_s :

- **No normalization**
 - input is $\vec{F}$ (Hz)
 - input is $\vec{G}$ (log Hz), allowing an assessment of whether log-transformation in and off itself helps.
- **Normalization with $\hat{\psi}_s$ estimated using the ‘classic approach’ in [Equation 6](#)**
 - over the entire training data (*PSTM1 (balanced data)*)
 - based on the individual token (*PSTM1 (single trial)*), as a more direct baseline for the remaining methods (all of which are based on individual tokens)
- **Normalization using the most commonly used normalization method (Lobanov 1971)**
 - using the entire training data (Hz). This method involves standardization of each speaker’s formants by subtracting the mean and dividing by the standard deviation of each formant, independently for each time point.
- **Normalization with $\hat{\psi}_s$ estimated using *PSTM***
 - method 2
 - method 3
 - method 6

8. References

- Assmann, Peter F, and William F Katz. 2000. “Time-Varying Spectral Change in the Vowels of Children and Adults.” *The Journal of the Acoustical Society of America* 108 (4): 1856–66.
- Barreda, Santiago. 2012. “Vowel Normalization and the Perception of Speaker Changes: An Exploration of the Contextual Tuning Hypothesis.” *The Journal of the Acoustical Society of America* 132 (5): 3453–64.
- . 2020. “Vowel Normalization as Perceptual Constancy.” *Language* 96 (2).
- . 2021. “Perceptual Validation of Vowel Normalization Methods for Variationist Research.” *Language Variation and Change*, 1–27.
- Barreda, Santiago, and Terrance M Nearey. 2012. “The Direct and Indirect Roles of Fundamental Frequency in Vowel Perception.” *The Journal of the Acoustical Society of America* 131 (1): 466–77.

———. 2018. “A Regression Approach to Vowel Normalization for Missing and Unbalanced Data.” *The Journal of the Acoustical Society of America* 144 (1): 500–520.

Charlton, Benjamin D, William AH Ellis, Rebecca Larkin, and W Tecumseh Fitch. 2012. “Perception of Size-Related Formant Information in Male Koalas (*Phascolarctos Cinereus*).” *Animal Cognition* 15: 999–1006.

Chernyak, Bronya R., Ann R. Bradlow, Joseph Keshet, and Matthew Goldrick. 2024. “A Perceptual Similarity Space for Speech Based on Self-Supervised Speech Representations.” *JASA* 155 (6): 3915–29.

Hillenbrand, James M, Michael J Clark, and Robert A Houde. 2000. “Some Effects of Duration on Vowel Recognition.” *The Journal of the Acoustical Society of America* 108 (6): 3013–22.

Hillenbrand, James M, Laura A Getty, Michael J Clark, and Kevin Wheeler. 1995. “Acoustic Characteristics of American English Vowels.” *The Journal of the Acoustical Society of America* 97 (5): 3099. <https://doi.org/10.1121/1.413041>.

Hillenbrand, James M, and Terrance M Nearey. 1999. “Identification of Resynthesized/hVd/Utterances: Effects of Formant Contour.” *The Journal of the Acoustical Society of America* 105 (6): 3509–23.

Jin, Zhengyang, Yuhao Zhu, and T. Florian Jaeger. 2025. “Latent Speech Representations Learned Through Self-Supervised Learning Predict Listeners’ Generalization of Adaptation Across Talkers.”

Johnson, K. 1997. “Speech Perception Without Speaker Normalization.” In *Talker Variability in Speech Processing*, edited by K Johnson and J W Mullennix, 145–46. San Diego: Academic Press.

Kim, Seung-Eun, Bronya R Chernyak, Olga Seleznova, Joseph Keshet, Matthew Goldrick, and Ann R Bradlow. 2024. “Automatic Recognition of Second Language Speech-in-Noise.” *JASA Express Letters* 4 (2).

Kleinschmidt, Dave F, and T Florian Jaeger. 2015. “Robust Speech Perception: Recognize the Familiar, Generalize to the Similar, and Adapt to the Novel.” *Psychological Review* 122 (2): 148.

Kleinschmidt, Dave F, Kodi Weatherholtz, and T Florian Jaeger. 2018. “Sociolinguistic Perception as Inference Under Uncertainty.” *Topics in Cognitive Science* 10 (4): 818–34.

Lobanov, Boris M. 1971. “Classification of Russian Vowels Spoken by Different Speakers.” *Journal of the Acoustical Society of America* 49 (2B): 606–8.

Luce, Paul A, and David B Pisoni. 1998. “Recognizing Spoken Words: The Neighborhood Activation Model.” *Ear and Hearing* 19 (1): 1–36.

- Magnuson, James S, and Howard C Nusbaum. 2007. "Acoustic Differences, Listener Expectations, and the Perceptual Accommodation of Talker Variability." *Journal of Experimental Psychology: Human Perception and Performance* 33 (2): 391.
- Massaro, Dominic W, and Daniel Friedman. 1990. "Models of Integration Given Multiple Sources of Information." *Psychological Review* 97 (2): 225.
- Nearey, T M, and Peter F Assmann. 2007. "Probabilistic 'Sliding-Template' Models for Indirect Vowel Normalization." *Experimental Approaches to Phonology* 246.
- Norris, Dennis, and James M McQueen. 2008. "Shortlist b: A Bayesian Model of Continuous Speech Recognition." *Psychological Review* 115 (2): 357.
- Persson, Anna, Santiago Barreda, and T Florian Jaeger. 2025. "Comparing Accounts of Formant Normalization Against US English Listeners' Vowel Perception." *Journal of the Acoustical Society of America* 157: 1458–82.
- Peterson, Gordon E., and Harold L. Barney. 1952. "Control Methods Used in a Study of the Vowels." *The Journal of the Acoustical Society of America* 24 (2): 175–84.
<https://doi.org/10.1121/1.1906875>.
- R Core Team. 2023. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Reby, David, Karen McComb, Bruno Cargnelutti, Chris Darwin, W Tecumseh Fitch, and Tim Clutton-Brock. 2005. "Red Deer Stags Use Formants as Assessment Cues During Intrasexual Agonistic Interactions." *Proceedings of the Royal Society B: Biological Sciences* 272 (1566): 941–47.
- Richter, Caitlin, Naomi H. Feldman, Harini Salgado, and Aren Jansen. 2017. "Evaluating Low-Level Speech Features Against Human Perceptual Data." *Transactions of the Association for Computational Linguistics* 5: 425–40.
https://doi.org/10.1162/tacL_a_00071.
- RStudio Team. 2024. *RStudio: Integrated Development Environment for r*. Boston, MA: RStudio, PBC. <http://www.rstudio.com/>.
- Sumner, Meghan. 2011. "The Role of Variation in the Perception of Accented Speech." *Cognition* 119 (1): 131–36.
- Taylor, AM, David Reby, and Karen McComb. 2010. "Size Communication in Domestic Dog, *Canis Familiaris*, Growls." *Animal Behaviour* 79 (1): 205–10.
- Xie, Xin, T Florian Jaeger, and Chigusa Kurumada. 2023. "What We Do (Not) Know about the Mechanisms Underlying Adaptive Speech Perception: A Computational Framework and Review." *Cortex* 166: 377–424.